- Inductors

$\underbrace{\Delta I_L}_{\text{ripple}} = \frac{V_{in} \cdot D}{L \cdot f}$ → Duty cycle $= 1 - \frac{V_{in}}{V_{out}}$

$L = \frac{3.7\left(1 - \frac{3.7}{5}\right)}{\Delta I_L \cdot f} = \frac{1.07 \times 10^{-6}}{\Delta I_L}$ ...(1)

↳ at 3.7 Vin, 0.9 MHz → target: 10% I_L for servo

$I_L \approx I_{in}$

$I_{in} \approx \frac{V_{out} \cdot I_{out}}{V_{in} \cdot \eta}$ ...(2)

V_{out} → 5V, V_{in} → 3.7V

(data sheet) η at $V_{in} = 3.7V \approx 80\%$ ..(3)

for servo boost, $I_{out} = 2 \times 750\ mA = 1.5A$ ...(4a)
for nano boost $I_{out} \approx 150\ mA$ (4b)

Hence, for servos, draws ≈ 2.5 A from battery

$L \approx 4.2\ \mu H$ ⇒ closest available $\approx 4.7\ \mu H$ ✓ → pros: quicker response to load change

Current can reach ~1.5A, so to be safe $I_{sat} > 2.5A$ ✓

for nano, draws ≈ 250 mA from battery

$L \approx 42.8\ \mu H$

BUT
- BIG Ahh inductor
- Slow response

Just make it same $4.7\ \mu H$, no matter

Later smoothered by decoupling capacitors $\Delta V \approx \frac{\Delta I}{8FC}$ → 100x smaller for nano compared to servo

↓ high

$I_{sat} > 1A$ enough.

conclusion:
- Servo boost $4.7\ \mu H$, $> 2.5A$
- Nano boost $4.7\ \mu H$, $> 1A$.

} shielded / molded to reduce EMI
Low DCR → datasheet MT 3608

- Resistors

Datasheet says:

$V_{out} = V_{REF} \times \left(1 + \frac{R_1}{R_2}\right) \Rightarrow \frac{5}{0.6} - 1 = \frac{R_1}{R_2} \Rightarrow \frac{R_1}{R_2} = 7.33$, $R_1 = 73.2\ k\Omega$, $R_2 = 10\ k\Omega$

$V_{REF} = 0.6$

common

- Capacitors

22 µF for input & output ceramic, low ESR, X5R / X7R } Vrating > 10V

decoupling use 100 nF and 220 µF just choose

- Diodes

datasheet says Schottky, current rating $= \sqrt{I_{out} \times I_{peak}} = \sqrt{1.5 \times 1.5}$, for safety > 2 A

RVB voltage > 5 V